Oral Exam 2 Information
REAL VARIABLES I
Fall 2026

Last updated: October 24, 2026.

0.1 Scheduling and exam format

Scheduling details and the appointment deadline will be announced.
When you arrive for the exam, you will be given a choice of two topics from the list below. You will select one topic and have up to 30 minutes to prepare your exposition. This preparation period is closed book: you may not use notes, books, or other resources.
After the preparation period, you will have an approximately 30-minute discussion at the board. You may consult any notes you produced during your preparation time.
During the board discussion, you should be prepared to:

- give a coherent overview of the selected topic and explain how its main ideas fit together;
- state definitions and mathematical statements precisely;
- present the central proofs at the level of their overall structure;
- supply detailed arguments for individual steps when asked; and
- respond to questions about hypotheses, examples, counterexamples, and connections among the results.

A presentation does not have to be flawless to earn full marks. The goal is to demonstrate that you have understood the topic and can think about it at the board. Tips or suggestions may be provided if you get stuck. If you are receptive to suggestions and recover from a temporary hesitation, you can still earn full marks.

0.2 Example grading rubric

The exam will be assessed holistically. The following descriptions illustrate the level of performance associated with each score range.

- **90–100%:** Demonstrates a strong and connected understanding of the topic. Definitions and statements are precise when requested, the main proof ideas are correct, and detailed arguments are supplied with only minor omissions or errors. Responds thoughtfully to questions and can use a tip to recover from hesitation. The presentation need not be perfect.
- **80–89%:** Demonstrates good understanding of the main ideas and proof structure. Most definitions and statements are accurate, but there are some gaps, imprecisions, or errors of substance. Responds productively to questions and suggestions, though some issues remain unresolved.
- **70–79%:** Demonstrates substantial but incomplete understanding. Can explain several central ideas and parts of the proofs, but important definitions, hypotheses, connections, or proof steps are missing or incorrect. Follow-up questions reveal uneven command of the topic. Can carry out at least one technical proof to completion.
- **60–69%:** Demonstrates limited understanding. Can state some relevant definitions or results and make partial progress on proofs, but the overall structure is unclear or contains errors. Substantial prompting is needed to proceed.

- **50–59%:** Demonstrates limited understanding. Some relevant facts are recalled, but central definitions, statements, and proof ideas are seriously incomplete or inaccurate. Makes little progress even with significant guidance.
- **Below 50%:** Does not yet demonstrate a workable understanding of the selected topic.

0.3 Topics

Status: The topic list is in progress. Additional topics will be added as the course progresses.
Unless an item is explicitly marked *statement only*, you should be able to give precise definitions and prove both the statement and auxiliary facts for the result. This applies to all mentioned facts: theorems, propositions, lemmata, or corollaries, and examples.

0.3.1 Topic 1: Comparison of the Riemann and Lebesgue integrals

1. **Definitions and distinctions.**

   - partitions, lower and upper sums $L(f;P)$ and $U(f;P)$, lower and upper Riemann integrals, and Riemann integrability of a bounded function on $[a,b]$;
   - the discontinuity set D_f and Lebesgue null sets;
   - lower and upper semicontinuity at a point and on $[a,b]$;
   - the lower and upper semicontinuous envelopes $\underline{f}$ and $\overline{f}$;
   - L-Lipschitz functions and the infimal convolution $f_n(x) = \inf_{y\in[a,b]}(f(y) + n|x-y|)$.

   Be able to explain why a lower sum is sensitive only to lower bounds on f, and an upper sum only to upper bounds, and how this leads to the two notions of semicontinuity.

2. **Lebesgue's criterion.** State precisely: a bounded $f\colon [a,b]\to\mathbb{R}$ is Riemann integrable if and only if $\mathcal{L}(D_f) = 0$, and in this case f is Lebesgue measurable and its Riemann and Lebesgue integrals agree. Explain the overall structure of the proof: express both Riemann integrals as Lebesgue integrals of the envelopes, then compare the envelopes.

3. **Semicontinuity through level sets.** Prove that g is lower semicontinuous if and only if every superlevel set $\{g > \alpha\}$ is open, and the analogous statement for upper semicontinuity. Deduce that semicontinuous functions are Borel measurable. Prove that the pointwise supremum of a family of lower semicontinuous functions is lower semicontinuous whenever it is finite.

4. **Semicontinuous envelopes.** Explain why $\underline{f}$ is the greatest lower semicontinuous function below f. Prove the local formula

$$\underline{f}(x) = \sup_{r>0}\inf_{y\in B_r(x)} f(y), \qquad \overline{f}(x) = \inf_{r>0}\sup_{y\in B_r(x)} f(y),$$

   and prove that f is continuous at x if and only if $\underline{f}(x) = \overline{f}(x)$. Deduce that D_f is a Borel set.

5. **Lipschitz approximation from below.** For a bounded lower semicontinuous f, prove that the infimal convolutions f_n are n-Lipschitz, satisfy $\inf f \le f_n \le f_{n+1} \le f$, and converge to f pointwise. Identify where lower semicontinuity is used. Be able to explain why replacing the linear penalty $n|x-y|$ by a hard cutoff (infimum over a ball) still gives approximants increasing to $\underline{f}$, but not continuous ones.

6. **Riemann integrals as integrals of the envelopes.** Prove

$$\underline{I}_R(f) = \int_{[a,b]} \underline{f}\, d\mathcal{L}, \qquad \overline{I}_R(f) = \int_{[a,b]} \overline{f}\, d\mathcal{L}.$$

   For the inequality $\leq$, construct a lower semicontinuous step function below f whose integral is $L(f;P)$. For the inequality $\geq$, compare the Lipschitz approximants of $\underline{f}$ with lower sums on fine uniform partitions and use monotone convergence. Explain how the upper identity follows from the lower one.

7. **Proof of Lebesgue's criterion.** Deduce the criterion from the previous item, using that a non-negative measurable function has zero integral if and only if it vanishes almost everywhere. Explain where completeness of Lebesgue measure is used to conclude that f is Lebesgue measurable, and why the two integrals agree.

8. **Examples and counterexamples.**

   - Compute the envelopes and the lower and upper Riemann integrals of the Dirichlet function $\mathbb{1}_{\mathbb{Q}\cap[0,1]}$; explain why it is Lebesgue integrable but not Riemann integrable.
   - Explain why boundedness cannot be dropped: $x^{-1/2}$ on $(0,1]$ is discontinuous at a single point, has no proper Riemann integral, and has Lebesgue integral 2.
   - Fat rationals: for an enumeration (q_k) of $\mathbb{Q}\cap[0,1]$ and $0 < r < 1/4$, let $F_n = \bigcup_{k=0}^{n} B_{r2^{-k}}(q_k)$ and $F = \bigcup_n F_n$. Prove that the Riemann integrable functions $\mathbb{1}_{F_n}$ increase to $\mathbb{1}_F$ and are Cauchy in the integral distance, and use Lebesgue's criterion to prove that $\mathbb{1}_F$ is not Riemann integrable. Explain what this example shows about monotone convergence and completeness for the Riemann integral.

Course notes: Review of Riemann Integration; The Lebesgue and Riemann Integrals.